

Heat Transfer Miscellany

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1 Thermal Diffusion Equation

Given a temperature gradient ∇T , the conductive heat flux is

$$\mathbf{F} = -\kappa \nabla T , \quad (1)$$

where κ is the thermal conductivity. If we imagine a bounded volume within the conductive medium, the rate of heat loss from the volume is

$$\text{Outward Power} = \oint_{\partial V} d\mathbf{A} \cdot \mathbf{F} = \int_V dV \nabla \cdot \mathbf{F} , \quad (2)$$

where the first integral is over the boundary surface and the second integral is over the interior volume. We now identify $\nabla \cdot \mathbf{F}$ as the outward conductive power per unit volume. Under conditions of constant pressure, which is not justified in general, the change in the heat content per unit mass is

$$dq = \left(\frac{\partial q}{\partial T} \right)_p dT \equiv c_p dT , \quad (3)$$

where c_p is the specific heat capacity per unit mass at constant pressure. Per unit volume of a material with density ρ , the rate of change of the heat content is $\rho c_p \partial_t T$, where we have adopted the shorthand $\partial_t \equiv \partial/\partial t$. If there are no sources of heat present and only conductive heat transport is operative, we have

$$\rho c_p \partial_t T = -\nabla \cdot \mathbf{F} = \nabla \cdot (\kappa \nabla T) . \quad (4)$$

A source (or sink) with power density S simply adds to the right-hand side of the above equation.

In what follows, we will assume that ρ , c_p , and κ are constant in space and time (a gross oversimplification), which allows us to write

$$\partial_t T - \alpha \nabla^2 T = S/\rho c_p , \quad (5)$$

where $\alpha = \kappa/\rho c_p$ is the constant thermal diffusivity. In the absence of sources, eq. (5) is a simple diffusion equation. Dimensional analysis of eq. (5) shows that in a time t diffusion progresses over a length scale $d \sim (\alpha t)^{1/2}$, the standard random-walk relation.

It is instructive to provide some quantitative perspective on the random-walk length by considering an actual material. Aluminum, for instance, has $\rho \simeq 2.7 \text{ g cm}^{-3}$, $c_p \simeq 0.9 \text{ J g}^{-1} \text{ K}^{-1}$, and $\kappa \simeq 2.5 \text{ J s}^{-1} \text{ cm}^{-1} \text{ K}^{-1}$, all taken at a temperature of $\simeq 300 \text{ K}$. Therefore, we find $\alpha \simeq 1 \text{ cm}^2 \text{ s}^{-1}$, so that in 1 s heat diffuses by approximately 1 cm in aluminum.

2 Green's Function

We can approach a more rigorous and general analysis by deriving the Green's function for the one-dimensional version of eq. (5). To this end, let $f(x, t) = S(x, t)/\rho c_p$, so that in 1D we have

$$\partial_t T - \alpha \partial_x^2 T = f(x, t) . \quad (6)$$

The Green's function $G(x - x'; t - t')$ is the solution of

$$\partial_t G - \alpha \partial_x^2 G = f_0 \delta(x - x') \delta(t - t') , \quad (7)$$

where f_0 is a constant. We see that G is the solution for an impulsive planar heat source. For a general source function, the solution involves a convolution with G :

$$T(x, t) = \int dx' \int dt' G(x - x'; t - t') [f(x', t')/f_0] , \quad (8)$$

which is easily verified by applying the differential operator $\partial_t - \alpha \partial_x^2$.

We can solve eq. (7) as follows. Let $\tilde{G}$ be the spatial Fourier transform of G :

$$G(x, t) = \frac{1}{\sqrt{2\pi}} \int dk e^{jkx} \tilde{G}(k, t) . \quad (9)$$

Also, for the spatial delta function we have

$$\delta(x - x') = \frac{1}{2\pi} \int dk e^{jk(x-x')} . \quad (10)$$

After making these substitutions, we obtain the complementary first-order differential equation

$$\partial_t \tilde{G} + \alpha k^2 \tilde{G} = \frac{f_0}{\sqrt{2\pi}} e^{-jkx'} \delta(t - t') . \quad (11)$$

This equation can be solved by first noting that $e^{-\alpha k^2 t}$ is a solution of the homogeneous equation (right-hand side equals zero) and then substituting $\tilde{G} = g(k, t) e^{-\alpha k^2 t}$ to obtain a simple equation for g :

$$\partial_t g = \frac{f_0}{\sqrt{2\pi}} e^{\alpha k^2 t} e^{-jkx'} \delta(t - t') , \quad (12)$$

and thus

$$g(k, t) = \frac{f_0}{\sqrt{2\pi}} e^{-jkx'} \int^t d\tau e^{\alpha k^2 \tau} \delta(\tau - t') . \quad (13)$$

If $t < t'$ the above integral vanishes, while when $t > t'$ the result is $e^{\alpha k^2 t'}$. Therefore, the solution for $\tilde{G}$ is

$$\tilde{G} = \frac{f_0}{\sqrt{2\pi}} e^{-jkx'} e^{-\alpha k^2 (t-t')} H(t - t') , \quad (14)$$

where $H(x)$ is the step function, which vanishes for $x < 0$ and equals 1 for $x > 0$. Computing the Fourier transform, we find

$$G(x - x'; t - t') = \frac{f_0 H(t - t')}{2\pi} \int dk \exp \{ -\alpha k^2 (t - t') + jk(x - x') \} . \quad (15)$$

Evaluation of the Gaussian integral yields

$$G(x - x'; t - t') = \frac{f_0 H(t - t')}{\sqrt{4\pi\alpha(t - t')}} \exp \left\{ -\frac{(x - x')^2}{4\alpha(t - t')} \right\} . \quad (16)$$

The Green's function is a spreading Gaussian for $t > t'$, where $\int dx G$ is a constant that represents the total energy per unit area delivered in the impulse. The spatial variance of the Gaussian is $2\alpha(t - t')$, which agrees with the dimensional analysis of the diffusion equation. The general solution to the heat equation with sources is now

$$T(x, t) = \int dx' \int^t dt' \frac{f(x', t')}{\sqrt{4\pi\alpha(t - t')}} \exp \left\{ -\frac{(x - x')^2}{4\alpha(t - t')} \right\} , \quad (17)$$

where we have used the step function to limit the t' integral.

3 Rectangular Heat Pulse

The Green's function solves the heat equation for an impulse of infinitesimal duration. We now consider a heat pulse of finite duration τ in the plane $x = 0$; the medium is assumed to be infinite. We take the functional form of the heat pulse to be

$$f(x, t) = f_0 \delta(x) [H(t) - H(t - \tau)] . \quad (18)$$

Equation (18) represents a rectangular step in time, such that the pulse is on for $0 < t < \tau$ and off otherwise. The general solution (eq. [17]) becomes

$$T(x, t) = f_0 \int_0^{\min(t, \tau)} dt' \frac{e^{-x^2/4\alpha(t-t')}}{\sqrt{4\pi\alpha(t-t')}} . \quad (19)$$

Under the change of variable $u^2 = x^2/4\alpha(t - t')$, we obtain

$$T(x, t) = \frac{f_0}{\sqrt{\pi}} \frac{x}{2\alpha} \int du u^{-2} e^{-u^2} \equiv \frac{f_0}{\sqrt{\pi}} \frac{x}{2\alpha} I , \quad (20)$$

where the integration limits have been suppressed for the time being. A single integration by parts yields

$$I = -u^{-1} e^{-u^2} - 2 \int du e^{-u^2} . \quad (21)$$

We see that the final result for $T(x, t)$ will contain an error function:

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z du e^{-u^2} . \quad (22)$$

At times $t \leq \tau$, while the pulse is on, the integral $I|_{u_0}^\infty$ is

$$I|_{u_0}^\infty = u_0^{-1} e^{-u_0^2} - \sqrt{\pi} [1 - \text{erf}(u_0)] , \quad (23)$$

where $u_0 = x/\sqrt{4\alpha t}$. After the pulse has turned off ($t > \tau$), the upper limit of the above integral is $u_\tau \equiv x/\sqrt{4\alpha(t - \tau)}$, so that

$$I|_{u_0}^{u_\tau} = u_0^{-1} e^{-u_0^2} - u_\tau^{-1} e^{-u_\tau^2} - \sqrt{\pi} [\text{erf}(u_\tau) - \text{erf}(u_0)] , \quad (24)$$

The temperature distributions for $t \leq \tau$ and $t > \tau$ are given by

$$T(x, t \leq \tau) = f_0 \left\{ \sqrt{\frac{t}{\pi\alpha}} e^{-x^2/4\alpha t} - \frac{x}{2\alpha} \left[1 - \operatorname{erf} \left(\frac{x}{\sqrt{4\alpha t}} \right) \right] \right\} \quad (25)$$

and

$$T(x, t > \tau) = f_0 \left\{ \sqrt{\frac{t}{\pi\alpha}} e^{-x^2/4\alpha t} - \sqrt{\frac{(t-\tau)}{\pi\alpha}} e^{-x^2/4\alpha(t-\tau)} + \frac{x}{2\alpha} \left[\operatorname{erf} \left(\frac{x}{\sqrt{4\alpha(t-\tau)}} \right) - \operatorname{erf} \left(\frac{x}{\sqrt{4\alpha t}} \right) \right] \right\} \quad (26)$$

Notice that at all times $T(x, t) \rightarrow 0$ as $x \rightarrow \infty$, as must be the case. If at time $t = 0$ the medium has a finite temperature T_0 for all x , then we simply add T_0 to the right-hand side of eqs. (25) and (26).

The heating of the slab during the “on” phase, as given by eq. (25), deserves further discussion. Let $z = x/\sqrt{4\alpha t}$, so that

$$T(x, t \leq \tau) = f_0 \sqrt{\frac{t}{\pi\alpha}} \left\{ e^{-z^2} - \sqrt{\pi} z [1 - \operatorname{erf}(z)] \right\}. \quad (27)$$

This form highlights the similarity of the solution that arises because there is no preferred length scale in the problem; length only appears in the dimensionless combination $x/\sqrt{\alpha t}$. The Green’s function (eq. [16]) also exhibits the same similarity property, as it must. In the limit of small z , corresponding to small x or large t , the linear approximation to eq. (27) is

$$T(z \ll 1, t) = f_0 \sqrt{\frac{t}{\pi\alpha}} (1 - \sqrt{\pi} z) = f_0 \left(\sqrt{\frac{t}{\pi\alpha}} - \frac{x}{2\alpha} \right). \quad (28)$$

As t increases, the linear approximation is valid for a wider range of x values. In effect, the temperature profile straightens out as energy is pumped into the slab, and the surface temperature increases in proportion to $t^{1/2}$.

Now perform the same analysis for times $t > \tau$, after the pulse has been turned off. Replacing the x dependence of eq. (26) with the similarity variable z , we find

$$T(x, t > \tau) = f_0 \sqrt{\frac{t}{\pi\alpha}} \left\{ e^{-z^2} - \sqrt{1 - \frac{\tau}{t}} e^{-z^2/(1-\tau/t)} + \sqrt{\pi} z \left[\operatorname{erf} \left(\frac{z}{\sqrt{1 - \tau/t}} \right) - \operatorname{erf}(z) \right] \right\}. \quad (29)$$

For small z , the $\mathcal{O}(z)$ term vanishes, and the surface temperature varies as

$$T(0, t) \propto t^{1/2} - (t - \tau)^{1/2}. \quad (30)$$

Letting $\Delta t = t - \tau$, we see that the surface temperature initially falls as $1 - (\Delta t/\tau)^{1/2}$ for $\Delta t/\tau \ll 1$, and ultimately transitions to the expected $\Delta t^{-1/2}$ dependence (see eq. [16]) when $t \gg \tau$. As t increases and the surface temperature decreases, the flattened portion of the temperature profile—the region where a quadratic approximation to $T(x)$ is acceptable—extends to larger x values.

4 Laser Heating

We can use the results of the last section, namely eqs. (25) and (26), to describe the heating of a semi-infinite medium in the half-space $x \geq 0$ by a laser pulse of finite duration and constant amplitude. Previously, we considered an infinite medium where heat was able to diffuse to both $x > 0$ and $x < 0$. For a semi-infinite medium, the deposited energy can diffuse only to $x > 0$ when we neglect convective and radiative surface cooling. From the definition of the rectangular pulse (eq. [18]) and upon referring to eq. (5), we see that $f_0 \rho c_p$ is the magnitude of the energy flux that gets transported to positive and negative x in an infinite medium. Therefore, the energy flux in the $+x$ direction is $f_0 \rho c_p / 2$. If F_l is the flux in the laser beam toward $x > 0$, then $f_0 = 2F_l / \rho c_p$. This can also be seen by differentiating eq. (28) with respect to x and recalling that eq. (1) defines the energy flux.

At the surface ($x = 0$), eq. (28) tells us that the temperature varies as

$$T(0, t \leq \tau) = T_0 + \frac{2F_l}{\rho c_p} \left(\frac{t}{\pi \alpha} \right)^{1/2} \quad (31)$$

in the “on” phase, and, from eq. (26),

$$T(0, t > \tau) = T_0 + \frac{2F_l}{\rho c_p} \left(\frac{t}{\pi \alpha} \right)^{1/2} \left[1 - \left(1 - \frac{\tau}{t} \right)^{1/2} \right] \quad (32)$$

after the pulse has turned off, where T_0 is the initial temperature. The bracketed term in eq. (32) causes the surface temperature to decay with time. The temperature rises while energy is being deposited, reaches a peak at time $t = \tau$, and then decays at times $t \gg \tau$.

While the pulse is on, the temperature increases within the medium according to eq. (27). The value of the similarity variable z at which the change in temperature is 1/2 of its surface value is $z_{1/2} \simeq 0.35$. This correct value of $z_{1/2}$ is not very different from the estimate obtained by extrapolating the linear approximation (eq. [28]; valid near $z = 0$), namely $z_{1/2}(\text{linear}) = (2\sqrt{\pi})^{-1} \simeq 0.28$. The position $x_{1/2}$ corresponding to $z_{1/2}$ is

$$x_{1/2} \simeq \sqrt{\alpha t / 2} , \quad (33)$$

consistent with the expected random-walk dependence $x \sim \sqrt{\alpha t}$.

A laser with 1 MW total power that is spread over a circular beam of radius 5–10 cm has a flux of order 10^4 W cm^{-2} . To incorporate this plausible scale, we write $F_l = 10^4 F_4 \text{ W cm}^{-2}$. Substituting the previously quoted values of ρ , c_p , and α for aluminum, we find for the peak temperature change at the surface

$$T(0, \tau) - T_0 = \frac{2F_l}{\rho c_p} \sqrt{\frac{\tau}{\pi \alpha}} \simeq 4600 F_4 \left(\frac{\tau}{1 \text{ s}} \right)^{1/2} \text{ K} . \quad (34)$$

We also obtain

$$x_{1/2} \simeq 0.7 \left(\frac{\tau}{1 \text{ s}} \right)^{1/2} \text{ cm} . \quad (35)$$

A pulse duration of 1 s and laser flux of $F_4 = 1$ is capable of raising the temperature of a 1 cm thick slab of aluminum to well over 1000 K. Such high temperatures can cause the aluminum to melt or to crack as a result of thermal stresses. It is important to note that this calculation assumes that

all of the laser power is deposited as heat into the material. In reality, only a fraction of the energy is absorbed.

The rather weak time dependence of the maximum surface temperature is an undesirable consequence of thermal diffusion. If the absorbed power is only 1% of the laser power, then the temperature increase might be only $\simeq 50$ K for $F_4 = 1$ and $\tau = 1$ s. In order for the temperature to reach 10^3 K, the pulse duration may need to be in excess of $[(1000 - T_0)/50]^2 \simeq 200$ s, which is unacceptably long.

5 Aerodynamic Heating

A very simple calculation tells us roughly the maximum temperature of the skin of an aircraft fuselage if the heating is due solely to aerodynamic friction. The no-slip condition very near the skin means that the bulk flow of air must come to a stop. If a fluid element moves along a streamline with constant entropy (i.e., the flow is adiabatic), its kinetic energy relative to the aircraft at some large distance upstream must be converted to heat near the fuselage.

For adiabatic flow, energy conservation along a streamline is represented by the Bernoulli relation¹,

$$h + v^2/2 = \text{constant} , \quad (36)$$

where $h = c_p T$ is the specific enthalpy of the gas. Suppose the aircraft moves at speed $v_a = M c_s$, where c_s is the sound speed of the ambient air and M is the Mach number. The change in temperature of air that approaches the skin is

$$\Delta T = \frac{M^2 c_s^2}{2 c_p} . \quad (37)$$

For an ideal gas, the sound speed is $c_s^2 = \gamma k T_\infty / m$, where T_a is the ambient air temperature, $\gamma = c_p / c_v$, and m is the mean molecular mass. From the thermodynamic relation $c_p = c_v + k/m$, we find

$$\frac{\gamma k}{m c_p} = \gamma - 1 , \quad (38)$$

so that

$$\Delta T = T_\infty \frac{(\gamma - 1)}{2} M^2 . \quad (39)$$

In air $\gamma - 1 \simeq 2/5$, and thus $\Delta T \simeq M^2/5$. If we set the ambient air temperature to $T_a = 300$ K, we expect that

$$\Delta T \simeq 60 M^2 \text{ K} . \quad (40)$$

For Mach numbers $\gtrsim 3$, the air temperature near the skin exceeds 500 K, which may introduce significant thermal stresses.

Our results indicate that a supersonic aircraft with $M \gtrsim 3$ is in danger of suffering structural fatigue or failure due to aerodynamic heating of the fuselage. Of course, the above temperature

¹The same quantity is conserved through a shock.

estimate is for the air near the skin, not the skin material itself. Heat from the air will be driven into the skin at a rate proportional to the temperature difference between the two media. However, we know, without calculating the heat flux, that the maximum possible skin temperature is given by eq. (40). Without a mechanism for actively cooling the aircraft shell, it seems likely that sustained supersonic flight with $M \gtrsim 3$ will introduce critically high temperatures.

Finally, we point out that an increase in the skin temperature of an aircraft will lessen the power or pulse duration of a high energy laser that intends to cause damage.